

Ex 14.6 To show: $z \in J(f_c) \Leftrightarrow -z \in J(f_c)$.

$$f_c(z) = z^2 + c$$

Since $f_c(-z) = f_c(z)$, $f_c^k(-z) = f_c^k(z)$

Hence, $z \in K(f_c) \Leftrightarrow -z \in K(f_c)$

and $z \in J(f_c) \Leftrightarrow -z \in J(f_c)$.

Ex 14.7 To show: If $c \in \mathbb{R}, c > \frac{1}{4}$, ~~the~~

$$z \notin J(f_c) \quad \forall z \in \mathbb{R}.$$

We'll show that $|f_c^k(z)| \rightarrow \infty$ as $k \rightarrow \infty$
 $\forall z \in \mathbb{R}$.

(This implies $z \notin K(f_c) \Rightarrow z \notin J(f_c)$)

Let $z \in \mathbb{R}$.

$$f_c(z) = z^2 + c$$

$$\Rightarrow f_c(z) - z = z^2 - z + c = \left(z - \frac{1}{2}\right)^2 + c - \frac{1}{4} \geq c - \frac{1}{4}$$

$$\Rightarrow f_c(z) \geq z + \left(c - \frac{1}{4}\right) \quad \text{--- (1)}$$

Also, $z \in \mathbb{R} \Rightarrow f_c(z) \in \mathbb{R}$

$$\therefore f_c(f_c(z)) \geq f_c(z) + \left(c - \frac{1}{4}\right) \quad [\text{From (1)}]$$

$$\geq z + 2\left(c - \frac{1}{4}\right)$$

By induction, $f_c^k(z) \geq z + k\left(c - \frac{1}{4}\right) \rightarrow \infty$ as $k \rightarrow \infty$

$$\Rightarrow z \notin K(f_c).$$

Since $J(f_c) \neq \emptyset$ and by the prev. exercise

$$z \in J(f_c) \Leftrightarrow -z \in J(f_c)$$

We conclude that $J(f_c)$ is not connected.

$\therefore c \notin M \quad \forall \text{ real number } c > \frac{1}{4}.$

Ex 14.9 To show: $\overline{B(0, \frac{1}{4})} \subseteq M.$

Let $c \in \mathbb{C}$ s.t. $|c| \leq \frac{1}{4}.$

$$\text{Then } |f_c(z)| = |z^2 + c| \leq |z|^2 + |c| \leq |z|^2 + \frac{1}{4} \leq \frac{1}{2} \quad \forall |z| \leq \frac{1}{2}$$

By induction, $|f_c^k(z)| \leq \frac{1}{2} \quad \forall |z| \leq \frac{1}{2}$

In particular, $|f_c^k(0)| \leq \frac{1}{2} \quad \forall k \in \mathbb{N}.$

$\Rightarrow c \in M.$

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Ex 14.10 Show that $\overline{B(-1, \frac{1}{20})} \subseteq M.$

Let $c \in \overline{B(-1, \frac{1}{20})}$

Then $|c+1| \leq \frac{1}{20}$

We'll show that $|f_c^k(0)| \rightarrow \infty$ as $k \rightarrow \infty.$

$$\begin{aligned} |f_c(f_c(z))| &= |(z^2+c)^2+c| \\ &= |z^4 + 2cz^2 + c^2 + c| \\ &\leq |z|^4 + 2|c||z|^2 + |c|^2 + |c| \end{aligned}$$

If $|z| \leq \frac{1}{10}$, then $|f_c^2(z)| \leq \frac{1}{10^4} + 2\left(1 + \frac{1}{20}\right) + \frac{1}{20} \cdot \frac{1}{20} < \frac{1}{10}$

By induction, $|f_c^{2k}(z)| \leq \frac{1}{10} \quad \forall k$ if $|z| \leq \frac{1}{10}.$

$\therefore |f_c^{2k}(0)| \leq \frac{1}{10} \quad \forall k \Rightarrow |f_c^k(0)| \rightarrow \infty$ as $k \rightarrow \infty$

$\Rightarrow c \in M.$

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Ex 14.11 Show that if $\varepsilon > 0$ and if $|z| \geq \max(2+\varepsilon, |c|)$,

then $|f_c(z)| \geq |z|(1+\varepsilon)$.

Deduce that if $|c| > 2$, then $c \notin M$.

$$\begin{aligned} |f_c(z)| &= |z^2 + c| \geq |z|^2 - |c| \\ &= |z| \left(|z| - \frac{|c|}{|z|} \right), \text{ if } z \neq 0 \\ &\geq |z|(2+\varepsilon-1) = |z|(1+\varepsilon) \\ &\text{for } |z| \geq \max(2+\varepsilon, |c|) \end{aligned}$$

Now, if $|c| > 2$, we can choose $\varepsilon > 0$ s.t.

$$|c| > 2+\varepsilon$$

$$\begin{aligned} \text{Then } |f_c(0)| &= |c| = \max(2+\varepsilon, |c|) \\ \therefore |f_c(f_c(0))| &\geq |f_c(0)|(1+\varepsilon) = |c|(1+\varepsilon) > |c| \end{aligned}$$

By induction,

$$|f_c^k(0)| \geq |c|(1+\varepsilon)^{k-1} \rightarrow \infty \text{ as } k \rightarrow \infty.$$

$$\Rightarrow c \notin M.$$

Ex 14.12 To show: $\mathbb{C} \setminus M = \{c \in \mathbb{C} : |f_c^k(0)| > 2 \text{ for some } k \in \mathbb{N}\}$

Suppose $|f_c^k(0)| > 2$ for some k .

We consider two cases:

Case (i): $|c| > 2$. Then $c \notin M$ by Ex 14.11

Case (ii): $|c| \leq 2$. Then $|c| < 2+\varepsilon \quad \forall \varepsilon > 0$

Since $|f_c^k(0)| > 2$, we choose $\varepsilon > 0$ s.t. $|f_c^k(0)| > 2+\varepsilon \geq \max(2+\varepsilon, |c|)$

Then $|f_c(f_c^k(o))| \geq |f_c^k(o)| (1+\epsilon) \quad (\text{by Ex 4.11})$
 $> 2+\epsilon$

By induction, $|f_c^n(f_c^k(o))| > (1+\epsilon)^n |f_c^k(o)|$
 $\rightarrow \infty \text{ as } n \rightarrow \infty.$

$\Rightarrow c \notin M.$

Conversely, if $c \notin M$, then

$|f_c^k(o)| \rightarrow \infty \text{ as } k \rightarrow \infty$

$\Rightarrow |f_c^k(o)| > 2 \text{ for some } k.$

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